

One Interval, Two Cutoffs

Unit 4 · Topic 4.3 · THE PROBLEM

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does the range 39.923 to 44.877 make both time claims believable? Give one reason.

DOK 1 (a) Locate 42 and 45 relative to the interval.

DOK 2 (b) Interpret the supplied 95% interval in one sentence about the mean setup time for all 800 transactions.

DOK 3 (c) In 3–4 sentences, decide whose approach is justified. Use the interval to judge each cutoff, then explain what Gabe gets wrong about individual transaction times.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.